over into Generative AI, exploring how neural networks can learn underlying data distributions to create entirely new, realistic content from scratch.

We will be unpacking the core architectures driving today's generative revolution:

- Autoencoders: We will start with the fundamentals of representation learning, examining how networks can compress data into dense latent spaces and reconstruct it, laying the conceptual groundwork for generative tasks.
- Generative Adversarial Networks (GANs): We will explore the adversarial framework—pitting a generator against a discriminator—that forces models to produce highly realistic synthetic data.
- Diffusion Models: You will learn the mechanics behind modern state-of-the-art image generators, breaking down how they synthesize new data by systematically reversing a noise-adding process.
- Text Generation: We will shift from visual data to natural language, exploring how sequence models and large language architectures predict and generate coherent, context-aware text.
- Speech Generation: Finally, we will look at how these generative principles extend to audio waveforms, enabling systems to synthesize natural, human-sounding speech.

By the end of this week, you will have a solid conceptual and practical grasp of the architectures powering the current wave of creative AI. Let's get started.

Learning Outcomes

- Deconstruct the architectural frameworks of modern generative models—including AutoEncoders, GANs, and Diffusion Models—to explain how neural networks learn underlying data distributions to synthesize realistic, novel visual data.